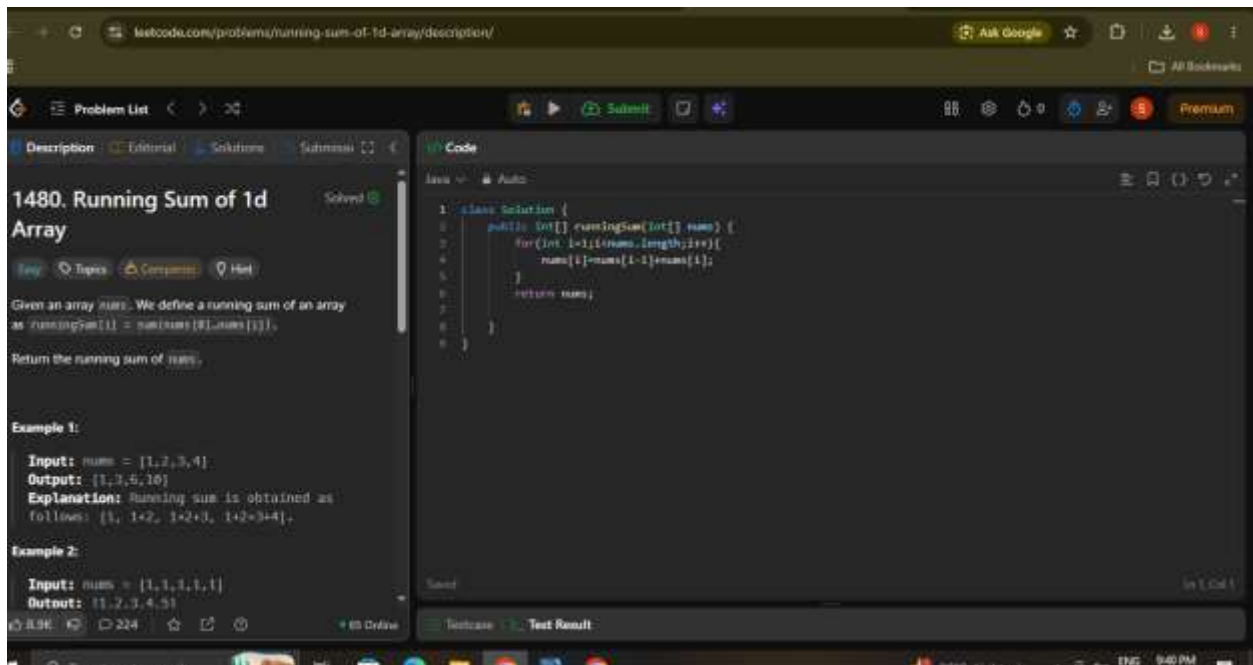


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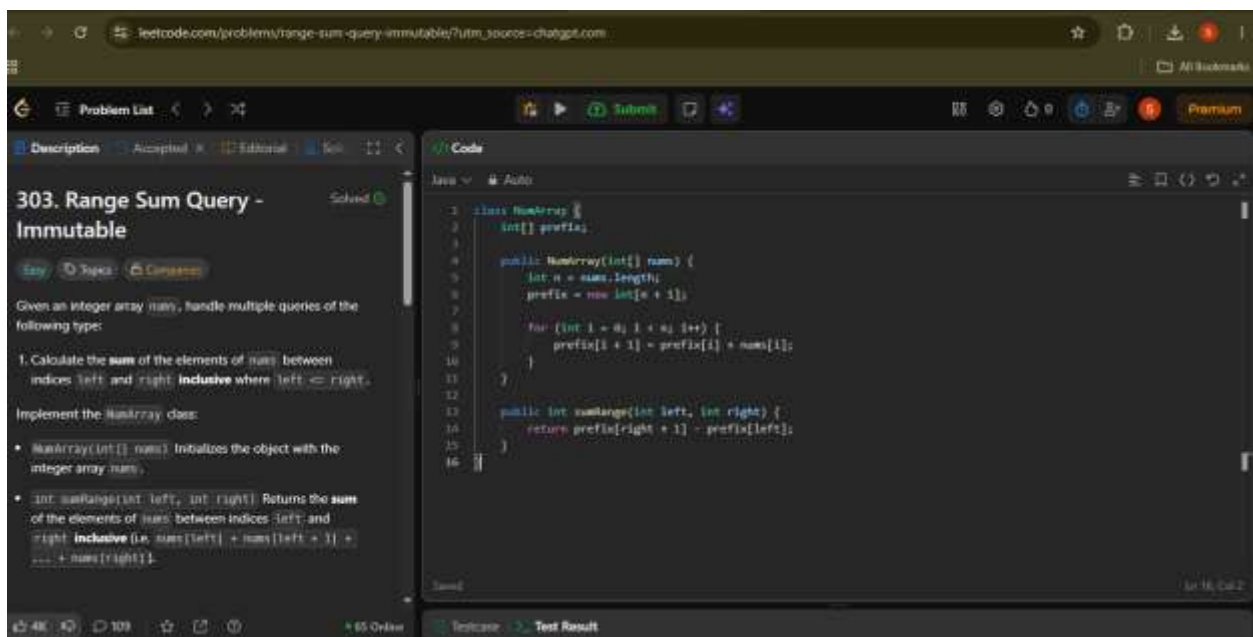
1480. Running Sum of 1d Array

Given an array `nums`. We define a running sum of an array as `runningSum[i] = sum(nums[0..i])`. Return the running sum of `nums`.

Example 1:
Input: `nums = [1,2,3,4]`
Output: `[1,3,6,10]`
Explanation: Running sum is obtained as follows: `[1, 1+2, 1+2+3, 1+2+3+4]`.

Example 2:
Input: `nums = [1,1,1,1,1]`
Output: `[1,2,3,4,5]`

```
1 class Solution {
2     public int[] runningSum(int[] nums) {
3         for (int i = 1; i < nums.length; i++) {
4             nums[i] = nums[i-1] + nums[i];
5         }
6         return nums;
7     }
8 }
```



303. Range Sum Query - Immutable

Given an integer array `nums`, handle multiple queries of the following type:

1. Calculate the sum of the elements of `nums` between indices `left` and `right` inclusive where `left <= right`.

Implement the `NumArray` class:

- `NumArray(int[] nums)` Initializes the object with the integer array `nums`.
- `int sumRange(int left, int right)` Returns the sum of the elements of `nums` between indices `left` and `right` inclusive (i.e. `nums[left] + nums[left + 1] + ... + nums[right]`).

```
1 class NumArray {
2     int[] prefix;
3
4     public NumArray(int[] nums) {
5         int n = nums.length;
6         prefix = new int[n + 1];
7
8         for (int i = 0; i < n; i++) {
9             prefix[i + 1] = prefix[i] + nums[i];
10        }
11    }
12
13    public int sumRange(int left, int right) {
14        return prefix[right + 1] - prefix[left];
15    }
16 }
```

Problem List

Description

Accepted

Solutions

1991. Find the Middle Index in Array

Solved

Easy

Topics

Companies

Hint

Given a 0-indexed integer array `nums`, find the **leftmost** `middleIndex` (i.e., the smallest amongst all the possible ones).

A `middleIndex` is an index where $nums[0] + nums[1] + \dots + nums[middleIndex-1] == nums[middleIndex+1] + \dots + nums[nums.length-1]$.

If `middleIndex` == 0, the left side sum is considered to be 0. Similarly, if `middleIndex` == `nums.length - 1`, the right side sum is considered to be 0.

Return the **leftmost** `middleIndex` that satisfies the condition, or `-1` if there is no such index.

Example 1:

1.6K 48 33 Online

Code

Java

Auto

```

1 class Solution {
2     public int findMiddleIndex(int[] nums) {
3         int total = 0;
4         for (int num : nums) {
5             total += num;
6         }
7         int leftsum = 0;
8
9         for (int i = 0; i < nums.length; i++) {
10
11             int rightsum = total - leftsum - nums[i];
12
13             if (leftsum == rightsum) {
14                 return i;
15             }
16
17             leftsum += nums[i];
18         }
19         return -1;
20     }
21 }

```

Speed

Testcase

Test Result

Problem List

Description

Editorial

Solutions

Submissions

724. Find Pivot Index

Solved

Easy

Topics

Companies

Hint

Given an array of integers `nums`, calculate the **pivot index** of this array.

The **pivot index** is the index where the sum of all the numbers **strictly** to the left of the index is equal to the sum of all the numbers **strictly** to the index's right.

If the index `n` is on the left edge of the array, then the left sum is 0 because there are no elements to the left. This also applies to the right edge of the array.

Return the **leftmost pivot index**. If no such index exists, return `-1`.

Example 1:

Input: `nums = [1,7,3,6,5,6]`

8.5K 278 119 Online

Code

Java

Auto

```

1 class Solution {
2     public int pivotIndex(int[] nums) {
3         int sum0;
4         for(int n:nums){
5             sum0+=n;
6         }
7         int leftsum=0;
8         for(int i=0;i<nums.length;i++){
9             int rightsum=sum-leftsum-nums[i];
10             if(rightsum==leftsum){
11                 return i;
12             }
13             leftsum+=nums[i];
14         }
15         return -1;
16     }
17 }

```

Speed

Testcase

Test Result

Problem List

560. Subarray Sum Equals K

Solved

Description

Editorial

Solutions

Submissions

Medium

Topics

Companies

Map

Given an array of integers `nums` and an integer `k`, return the total number of subarrays whose sum equals to `k`.

A subarray is a contiguous **non-empty** sequence of elements within an array.

Example 1:

Input: `nums = [1,1,1], k = 2`
Output: 2

Example 2:

Input: `nums = [1,2,3], k = 3`
Output: 2

25.7k 438 482 Doves

Code

Java

Auto

```
1 class Solution {
2     public int subarraySum(int[] nums, int k) {
3         HashMap<Integer,Integer>map=new HashMap<>();
4         map.put(0,1);
5         int sum=0;
6         int count=0;
7         for(int i=0;i<nums.length;i++){
8             sum+=nums[i];
9             if(map.containsKey(sum-k)){
10                 count+=map.get(sum-k);
11             }
12             map.put(sum,map.getOrDefault(sum,0)+1);
13         }
14         return count;
15     }
16 }
```

Testcase Test Result

Problem List

525. Contiguous Array

Solved

Description

Editorial

Solutions

Submissions

Medium

Topics

Companies

Given a binary array `nums`, return the maximum length of a contiguous subarray with an equal number of 0 and 1.

Example 1:

Input: `nums = [0,1]`
Output: 2
Explanation: [0, 1] is the longest contiguous subarray with an equal number of 0 and 1.

Example 2:

Input: `nums = [0,1,0]`
Output: 2
Explanation: [0, 1] (or [1, 0]) is a longest contiguous subarray with equal number of 0 and 1.

Code

Java

Auto

```
1 class Solution {
2     public int findMaxLength(int[] nums) {
3         int length=0;
4         int sum=0;
5         HashMap<Integer,Integer>map=new HashMap<>();
6         map.put(0,1);
7         for(int i=0;i<nums.length;i++){
8             sum+=(nums[i]==0)?-1:1;
9             if(map.containsKey(sum)){
10                 length=Math.max(length,i-map.get(sum));
11             }
12             else{
13                 map.put(sum,i);
14             }
15         }
16         return length;
17     }
18 }
```

Testcase Test Result

Problem List

704. Binary Search

Solved

Tags

Topics

Companies

Given an array of integers `nums`, which is sorted in ascending order, and an integer `target`, write a function to search `target` in `nums`. If `target` exists, then return its index. Otherwise, return `-1`.

You must write an algorithm with $O(\log n)$ runtime complexity.

Example 1:

Input: `nums = [-1, 0, 3, 5, 9, 12]`, `target = 9`
Output: `4`
Explanation: 9 exists in `nums` and its index is 4.

Example 2:

13.7K

230

255 Online

Testcase

Test Result

Auto

```

1 class Solution {
2     public int search(int[] nums, int target) {
3         int low=0;
4         int high=nums.length-1;
5         while(low<high){
6             int mid=low+(high-low)/2;
7             if(nums[mid]==target){
8                 return mid;
9             }
10            else if(nums[mid]<target){
11                low=mid+1;
12            }
13            else{
14                high=mid;
15            }
16        }
17        return -1;
18    }
19 }

```

Save

10 / 100

Problem List

35. Search Insert Position

Solved

Tags

Topics

Companies

Given a sorted array of distinct integers and a target value, return the index if the target is found. If not, return the index where it would be if it were inserted in order.

You must write an algorithm with $O(\log n)$ runtime complexity.

Example 1:

Input: `nums = [1, 3, 5, 6]`, `target = 5`
Output: `2`

Example 2:

Input: `nums = [1, 3, 5, 6]`, `target = 2`
Output: `1`

10.1K

478

222 Online

Testcase

Test Result

Auto

```

1 class Solution {
2     public int searchInsert(int[] nums, int target) {
3         int low=0;
4         int high=nums.length-1;
5         while(low<high){
6             int mid=low+(high-low)/2;
7             if(nums[mid]==target){
8                 return mid;
9             }
10            else if(nums[mid]<target){
11                low=mid+1;
12            }
13            else{
14                high=mid;
15            }
16        }
17        return low;
18    }
19 }

```

Recover from local

Upgrade to Cloud Saving

10 / 100

Problem List

69. Sqrt(x) Solved

Easy Topics Companies

Given a non-negative integer `x`, return the square root of `x` rounded down to the nearest integer. The returned integer should be **non-negative** as well.
You must not use any built-in exponent function or operator.
• For example, do not use `pow(x, 0.5)` in c++ or `x ** 0.5` in python.
Example 1:
Input: `x = 4`
Output: `2`
Explanation: The square root of 4 is 2, so we return 2.
Example 2:

Code

Java Auto

```
1 class Solution {
2     public int mySqrt(int x) {
3         if(x==0||x==1)return x;
4         int low=1;
5         int end=x;
6         while(low<end){
7             int mid=low+(end-low)/2;
8             if(mid==x/mid){
9                 return mid;
10            }
11            else if(mid<x/mid){
12                low=mid+1;
13            }
14            else{
15                end=mid-1;
16            }
17        }
18        return end;
19    }
20 }
```

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9.9K 481 155 Online Testcase Test Result

Problem List

34. Find First and Last Position of Element in Sorted Array Solved

Medium Topics Companies

Given an array of integers `nums` sorted in non-decreasing order, find the starting and ending position of a given `target` value.
If `target` is not found in the array, return `[-1, -1]`.
You must write an algorithm with $O(\log n)$ runtime complexity.
Example 1:
Input: `nums = [5,7,7,8,8,10], target = 8`
Output: `[3,4]`
Example 2:

Code

Java Auto

```
1 class Solution {
2     public int[] searchRange(int[] nums, int target) {
3         int first=search1(nums,target);
4         int second=search2(nums,target);
5         return new int[]{first,second};
6     }
7     int search(int nums[],int target){
8         int low=0;
9         int high=nums.length-1;
10        int ans=-1;
11        while(low<high){
12            int mid=low+(high-low)/2;
13            if(nums[mid]==target){
14                ans=mid;
15                low=mid+1;
16            }
17            else if(nums[mid]<target){
18                low=mid+1;
19            }
20            else{
21                high=mid-1;
22            }
23        }
24        return ans;
25    }
26 }
```

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23.6K 397 238 Online Testcase Test Result

Problem List

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⚙️

85

Description

Editorial

Solutions

Submissions

33. Search in Rotated Sorted Array

Solved

Medium

Topics

Companies

There is an integer array `nums` sorted in ascending order (with distinct values).

Prior to being passed to your function, `nums` is **possibly left rotated** at an unknown index k ($1 \leq k < \text{nums.length}$) such that the resulting array is `[nums[k], nums[k+1], ..., nums[n-1], nums[0], nums[1], ..., nums[k-1]]` (**0-indexed**). For example, `[0,1,2,4,5,6,7]` might be left rotated by `3` indices and become `[4,5,6,7,0,1,2]`.

Given the array `nums` after the possible rotation and an integer `target`, return the index of `target` if it is in `nums`, or `-1` if it is not in `nums`.

You must write an algorithm with $O(\log n)$ runtime complexity.

30.7K

716

12

505 Online

Code

Java

Auto

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Test Result

Problem List

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⚙️

85

Description

Editorial

Solutions

Submissions

162. Find Peak Element

Solved

Medium

Topics

Companies

A peak element is an element that is strictly greater than its neighbors.

Given a **0-indexed** integer array `nums`, find a peak element, and return its index. If the array contains multiple peaks, return the index to **any of the peaks**.

You may imagine that `nums[-1] = nums[n] = -∞`. In other words, an element is always considered to be strictly greater than a neighbor that is outside the array.

You must write an algorithm that runs in $O(\log n)$ time.

Example 1:

Input: `nums = [1,2,3,1]`

Output: `2`

14.9K

566

12

185 Online

Code

Java

Auto

Save

Testcase

Test Result

The screenshot shows the LeetCode problem page for "74. Search a 2D Matrix". The problem description states: "You are given an $m \times n$ integer matrix `matrix` with the following two properties:

- Each row is sorted in non-decreasing order.
- The first integer of each row is greater than the last integer of the previous row.

Given an integer `target`, return `true` if `target` is in `matrix` or `false` otherwise.

You must write a solution in $O(\log m + \log n)$ time complexity.

Example 1:

1	3	5	7
---	---	---	---

The right side of the image shows the C++ code for the solution, which uses a binary search approach across rows and columns.

```

class Solution {
public:
    bool searchMatrix(vector<vector<int>> &matrix, int target) {
        int m = matrix.size();
        int n = matrix[0].size();
        int s = 0;
        int e = m - 1;
        while(s <= e){
            int mid = (s + e) / 2;
            int row = mid % n;
            int col = mid / n;
            if(matrix[row][col] == target){
                return true;
            } else if(matrix[row][col] < target){
                s = mid + 1;
            } else{
                e = mid - 1;
            }
        }
        return false;
    }
};

```

Problem List

875. Koko Eating Bananas

Medium

Koko loves to eat bananas. There are n piles of bananas, the i^{th} pile has $piles[i]$ bananas. The guards have gone and will come back in h hours.

Koko can decide her bananas-per-hour eating speed of k . Each hour, she chooses some pile of bananas and eats k bananas from that pile. If the pile has less than k bananas, she eats all of them instead and will not eat any more bananas during this hour.

Koko likes to eat slowly but still wants to finish eating all the bananas before the guards return.

Return the minimum integer k such that she can eat all the bananas within h hours.

Example 1:

```
Input: piles = [3,6,7,4], h = 8
Output: 4
```

Code

```

1 class Solution {
2     public int minEatingSpeed(int[] piles, int h) {
3         int low=1;
4         int high=0;
5         for(int n:piles){
6             high=Math.max(high,n);
7         }
8         int ans=high;
9         while(low<high){
10            int mid=low+(high-low)/2;
11            if(isPossible(piles,h,mid)){
12                ans=mid;
13                high=mid-1;
14            }
15            else{
16                low=mid+1;
17            }
18        }
19        return ans;
20    }
21 }
```

34.2K
887
311 Online

[Testcase](#)
[Test Result](#)